15/02/24 FN

Hall Ticket Number:

MR-22

Code: B324CS5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular Examinations, Jan/Feb-2024

Computer Organization and Architecture

(Common to CSE, CSB, CSM & CSD)

Time: 3 Hours

Answer ALL Questions

Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Draw the block diagram to depict P: R2←R1.	1	1	1	1
1. b)	What is the role of PC and IR?	1	1	2	1
1. c)	What is the role of micro programmed control unit?	1	2	2	2
1. d)	What is the difference between ADD and ADDC instructions?	1	2	2	2
1. e)	Convert the following into binary numbers. i. $(ACD)_{16}=(_____)_2$ ii. $(657)_8=(_____)_2$	1	3	5	2
1. f)	Draw the hardware configuration for signed-magnitude addition and subtraction.	1	1	5	1
1. g)	What is pipelining?	1	1	3	1
1. h)	Define mapping in cache memory?	1	1	3	1
1. i)	List the characteristics of CISC architecture.	1	1	3	2
1. j)	Write the classification of parallel processing.	1	1	4	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	The initial values of 8-bit registers R1, R2, R3, R4 are given below. R1=11110000, R2=11111111, R3=10111001, R4=11101010 Determine the values of the registers R1, R2, R3, R4 after execution of the following operations. R1←R1+R2 R3←R3^R4 R2←R2+1 R1←R1-R3	5	3	2	1
2. b)	List and explain the phases of instruction cycle with flowchart.	5	2	2	1
	OR				
2. c)	Consider the register R=11011101. Determine the value of R after performing the following operations in sequence. i. Logical shift left ii. Circular shift right iii. Logical shift right iv. Circular shift left	5	3	2	1
2. d)	Draw the common bus structure to connect basic computer registers.	5	2	2	1
3. a)	Differentiate hardwired control and microprogrammed control.	5	2	2	3
3. b)	Write the assembly language program to evaluate A = (B+C)/D*E using two address instruction format.	5	3	2	1
	OR				
3. c)	Explain the fields of microinstruction instruction format.	5	2	2	1
3. d)	Explain different addressing modes present in accessing the data in memory?	5	3	2	1

P.T.O.

4. a)	Perform the subtraction with the following unsigned binary numbers. i. 11010-1000 ii. 11010-1101 iii. 10101-101 iv. 1111-0101	5	3	5	2
4. b)	Draw the flow chart for decimal division and explain division of 23 with 4.	5	3	5	2
	OR				
4. c)	Draw the flow chart of floating point addition operation?	5	3	5	1
4. d)	Draw the flow chart for booth multiplication and explain multiplication of 12 with 13.	5	3	5	3
5. a)	Explain the working of DMA controller.	5	2	3	1
5. b)	A cache consists of memory size of 256X16 and 8 blocks. The main memory contains memory size of 4096X16 .find out how many bits are there in Index, Tag, block, and Word fields in the case of direct mapping.	5	3	3	2
	OR				
5. c)	Discuss daisy-chain priority interrupt using neat diagram.	5	1	3	1
5. d)	Draw the classification diagram of semiconductor memory and explain briefly about ROM, PROM, EEPROM and Flash memory.	5	2	3	1
6. a)	Explain in detail about RISC instruction pipeline with data conflicts problem?	5	2	4	1
6. b)	Discuss about the following Interconnection structures i. Time-shared common bus ii. Crossbar switch	5	2	4	1
	OR				
6. c)	Explain about vector processing?	5	2	4	1
6. d)	What is cache coherence problem? Suggest the solution for it.	5	2	4	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome